Using Bilingual Responses to Enhance Cultural Knowledge in AI

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I. Project Details

- CODE is AVAILABLE
  • Limitation: BLEnD identifies that LLMs perform better in the local language for high-resource cultures and better in English for low-resource cultures, but it only analyzes this phenomenon and does not provide a mechanism to leverage it for improving answer accuracy. Additionally, it does not use any external cultural knowledge source to validate or guide response selection.
  • Proposed Solution: We propose a dual-language ensemble framework that generates answers in both the local language and English. An LLM-as-Judge then evaluates these responses and selects or synthesizes the best answer. To improve decision-making, the judge is further supported by a cultural knowledge base that serves as a reference for verifying cultural facts and resolving ambiguities, enabling more accurate and culturally grounded responses.

Team Composition and Individual Contributions:

- Member 1: Aarush Ranjan, Roll No. B24DS002, arushr@iitbhilai.ac.in, Phase 2: knowledge base construction (Wikipedia scraping, embeddings, FAISS index), dual-language generation, disagreement check, LLM-as-Judge and basic app design.
- Member 2: Joshitha Reddy Bangaru, Roll No. B24DS009, joshithaba@iitbhilai.ac.in, Phase 3: fixes and integration (linking the knowledge base to the API, single API server, evaluation table in the web application, latency test, input formats), running all experiments, the report and the PPT

II. Problem Statement

Aim of the Project: The aim of this project is to improve the accuracy of LLMs on cultural questions from different countries and languages. We do this by combining answers from both the local language and English, and then using an LLM-as-Judge with a cultural knowledge base to choose the best final answer **if the similarity is less than 0.75, which we set as a threshold**.

• Context and Motivation: Large Language Models (LLMs) are now used by people all over the world. However, previous work [**?**] showed that these models often make mistakes when answering simple cultural questions about countries that have less data available online. For example, questions about local foods, school traditions, festivals, family customs, or daily routines may receive generic or culturally incorrect answers. This happens because many cultures and languages are not well represented in the data used to train LLMs. The BLEnD benchmark found an interesting pattern: for some countries, LLMs answer better in the local language. For other countries, LLMs answer better in English. However, BLEnD only reports this observation and does not use it to improve answers. Following this idea, and based on our professor's suggestion, we add a cultural knowledge base that can act as an additional source of information during answer selection. This knowledge base helps the judge model verify cultural facts and make better decisions when answers from the two languages differ.

• Why this problem is important in the ML domain: This problem is important because LLMs are expected to help users from many different countries and language backgrounds. As AI becomes more widely used around the world, understanding different cultures correctly becomes an important ML challenge.

• Focus on work (sub-problem): Our project focuses on improving cultural question answering using a dual-language ensemble and knowledge-base-assisted adjudication. For each question: the LLM generates an answer in the local language. The same question is also answered in English. The two answers are compared. Relevant information is retrieved from a cultural knowledge base. An LLM-as-Judge uses both answers and the retrieved knowledge to select or create the final answer. Through this approach, we aim to reduce cultural knowledge gaps and improve answer accuracy across different countries and languages. In Phase 3 the pipeline was completed with an evaluation record for every question, a web application, a latency measurement for each stage and an ablation study that separates the effect of the knowledge base from the effect of the judge.

III. Methodology

• PROPOSED ML MODELS / ARCHITECTURE: MODELS: Multilingual LLM: generates answers in both English and the local language. Dual-Language Ensemble: treats the two language responses as complementary candidates. RAG (Retrieval-Augmented Generation): retrieves relevant cultural knowledge from the Knowledge Base. LLM as a Judge: compares the two answers using the retrieved evidence and selects/improves the final answer. Evidence-Grounded Answer Selection: the final answer must be supported by retrieved cultural knowledge. Overall architecture: Question → English LLM + Local Language LLM → Answer Comparison → Knowledge Base Retrieval → LLM Judge → Final Answer

• KEY TECHNIQUES, ALGORITHMS AND FRAMEWORKS: Key techniques: Multilingual prompting, Dual-language inference, RAG, Semantic similarity, Knowledge-base retrieval, Evidence-based verification, LLM-as-Judge, Answer selection/synthesis. Generate answers: Multilingual

LLM (Gemini API). Create text embeddings: Sentence Embeddings (`sentence-transformers`; multilingual-e5-base for answer similarity and multilingual-e5-large for the knowledge base) [**?**]. Find relevant knowledge from the Knowledge Base: Vector Similarity Search (FAISS) [**?**]. Compare the English and local-language answers: Cosine Similarity of the two answer embeddings. Detect disagreement between answers: the answers are treated as disagreeing when their cosine similarity is below the threshold of 0.75 (for multiple-choice questions, when the chosen option letters differ). Retrieve relevant evidence from the Knowledge Base: Top-3 retrieval (FAISS) restricted to the question's country. Make the final decision: LLM-as-Judge [**?**]. Evaluate answer accuracy: accuracy against the BLEnD reference answers, with the model's own assessment used only when no reference answer exists. Compare performance across languages: Accuracy / Performance Gap (numpy, pandas). No model training/fine-tuning is necessary for the core proposed system. The main novelty is the inference-time architecture.

• IMPLEMENTATION AND EVALUATION (PHASE 3): The system is a FastAPI service with a Streamlit web application. For every question the English and the local-language answers are produced by two separate model calls, so neither answer can influence the other; for multiple-choice questions both answers start with an option letter and different letters count as disagreement. When the answers disagree, three facts are retrieved from the knowledge base of the question's country and given to the judge, which returns the final answer and states whether it sided with the English answer, the local answer or neither. The pipeline is evaluated in five variants: English only, local only, judge without knowledge base, judge with knowledge base (the main system) and judge with knowledge base on every question. All model responses are cached, so the variants are compared on the same generated answers. Each question is stored as one record with the fields of Tables **??** and **??**. *Final Correct* is decided against the BLEnD reference answer (option letter for multiple choice; a whole-word match, followed by a model check against the reference, for short answers). Only when no reference answer exists does the model assess the answer itself, and such records are labelled and counted separately. *KB Grounded* is a model check of whether the retrieved facts support the final answer, and latency is recorded for each stage (generation, similarity, retrieval, judge). Accuracy is reported with 95% bootstrap confidence intervals and an exact McNemar test between the main system and the other variants.

• HOW OUR APPROACH DIFFERS FROM BLEnD: BLEnD is primarily a benchmark/evaluation framework: it measures cultural knowledge and shows that language choice affects performance. It does not use the local-vs-English observation as an answer-improvement mechanism. BLEnD: Local language vs English → measure which performs better. Our model: Local language + English → compare both → retrieve cultural evidence → judge → select/synthesize the better answer. So the key difference is: BLEnD diagnoses the cultural/language gap; our system actively uses that gap to improve the answer. The Knowledge Base adds an external cultural evidence source, reducing dependence on whichever language happens to produce the better answer.

• FLOWCHART (Fig. 1): BLEnD Question → English Prompt + Local Language Prompt → Multilingual LLM → English Answer + Local Answer → Answer Comparison / Disagreement Detection → Retrieve relevant Cultural Knowledge from KB (RAG) → LLM-as-Judge + Both Answers + Retrieved Evidence → Select / Synthesize Final Answer → Grading, Evaluation Record Table and Language/Culture Gap Analysis

IV. Resource and Dataset Availability

Provide details of the dataset(s) you will use if it is available:

- BLEnD dataset, Source of dataset: GitHub and Article shared Link: https://github.com/nlee0212/BLEnD, https://huggingface.co/datasets/nayeon212/BLEnD
- 5 countries (South Korea, Ethiopia, China, Indonesia and Azerbaijan) out of the real data of 52.6K QA pairs spanning 16 countries/regions. The evaluation uses a curated set of 50 short-answer questions (10 per country, with reference answers checked by us) and BLEnD questions taken in dataset order (**[FILL: number]** per country). The curated set is split into two equal parts of 5 questions per country, which are run separately.
- Each QA pair contains: Question: cultural knowledge question. Local-language version: question in the respective local language. English version: parallel English question. Answer: short-answer or multiple-choice format. Human vote annotations: human-annotated answers for evaluation. Formats: Short-answer QA and Multiple-choice QA, with parallel local/English versions. The reference answers for short-answer questions are the English annotator answers with at least two votes.
- For five selected countries/regions (South Korea, Ethiopia, Azerbaijan, Indonesia, and China) chosen to span high-, mid-, and low-resource languages, we query each test question twice per model: once using BLEnD's local-language version, once using its parallel English version.

List all resources you anticipate needing:

- Hardware: NVIDIA GPU with 8–16 GB VRAM (optional, since the LLM runs through an API), RAM: 16 GB, Storage: 20–30 GB free SSD space, Internet: required for LLM API calls and external Knowledge Base/API access.
- Software: Python 3.10+, FastAPI, Uvicorn, Streamlit, SQLite, sentence-transformers, FAISS (CPU), pandas, NumPy, matplotlib, Hugging Face `datasets`, and `wikipedia-api`.
- Additional tools and APIs: Google AI Studio (Gemini API) and the Wikipedia API for the knowledge base.

The Gemini API has a strict request limit, so the evaluation is run in parts over several days. Every model response is cached, failed calls are retried and never stored as answers, and several API keys (one per team member) are rotated automatically when a key reaches its limit.

TABLE I
MODEL RESULTS (THRESHOLD = 0.75) — PART 1: DETAILS

Record ID	Country/Region	Cultural Category	Question
6	South Korea	Clothing	What is the traditional Korean clothing worn on holidays?
7	South Korea	Food	What soup do Koreans eat on Seollal, the Lunar New Year'
5	Azerbaijan	Food	What foods are prepared for Novruz?
4	Indonesia	Festival/Holiday	What is Nyepi and what happens on that day?
3	China	Festival/Holiday	Why do people eat mooncakes, and when?
2	Ethiopia	Festival/Holiday	What is Meskel and how is it celebrated?
1	South Korea	Festival/Holiday	What is the cultural significance of Chuseok?

V. LIMITATIONS FOUND

BLEnD shows that LLM performance can vary between local-language and English responses, but does not use this difference to improve the final answer. Standard implementations usually generate an answer in one language and directly return it without cross-language verification. Our approach generates both local-language and English answers and uses their disagreement as a signal for further verification. It additionally uses a cultural Knowledge Base + LLM-as-Judge to make the final decision using external evidence. Thus, it combines dual-language inference, retrieval, and evidence-based adjudication rather than relying on a single model response.

- Difference from Standard Implementations: Standard: Question → LLM → Answer. Our approach: Question → Local + English LLM → Compare → Retrieve Cultural Knowledge → LLM-as-Judge → Final Answer.
- Improvements Aimed: Performance: Improve cultural QA accuracy, especially for low-resource languages/cultures. Efficiency: Avoid unnecessary adjudication when both language answers agree; retrieve only relevant KB evidence. Generalization: Improve robustness across languages, countries, question wording, and cultural contexts. Reliability: Reduce incorrect or generic answers by grounding the final decision in retrieved cultural evidence.
- Known limitations of this work: the knowledge base is built only from English Wikipedia, so everyday practices such as school routines or family customs are covered only partly; evidence is retrieved only when the answers disagree, so answers on which both languages agree but are wrong are not checked; the disagreement threshold of 0.75 was fixed in advance and not tuned; the model's own assessment of correctness, used when no reference answer exists, is less reliable than comparison with a reference; and results come from a limited number of questions and one model family.

VI. EXPECTED OUTCOMES

The deliverables and goals of the project are listed below. Every question is recorded with the fields of Tables **??** and **??**, the deliverables are summarised in Table **??** and the measured results in Table **??**.

APPENDIX

LLM (Claude) was used through its chat interface for coding assistance the pipeline fixes, the web application and the evaluation scripts, error debugging, improving the app options and drafting the LaTeX of this report. The team independently reviewed and approved the methodology, experiments, implementation, and final content. Tools used: Google AI Studio (Gemini Flash), Claude, FastAPI, Streamlit, FAISS and GitHub.

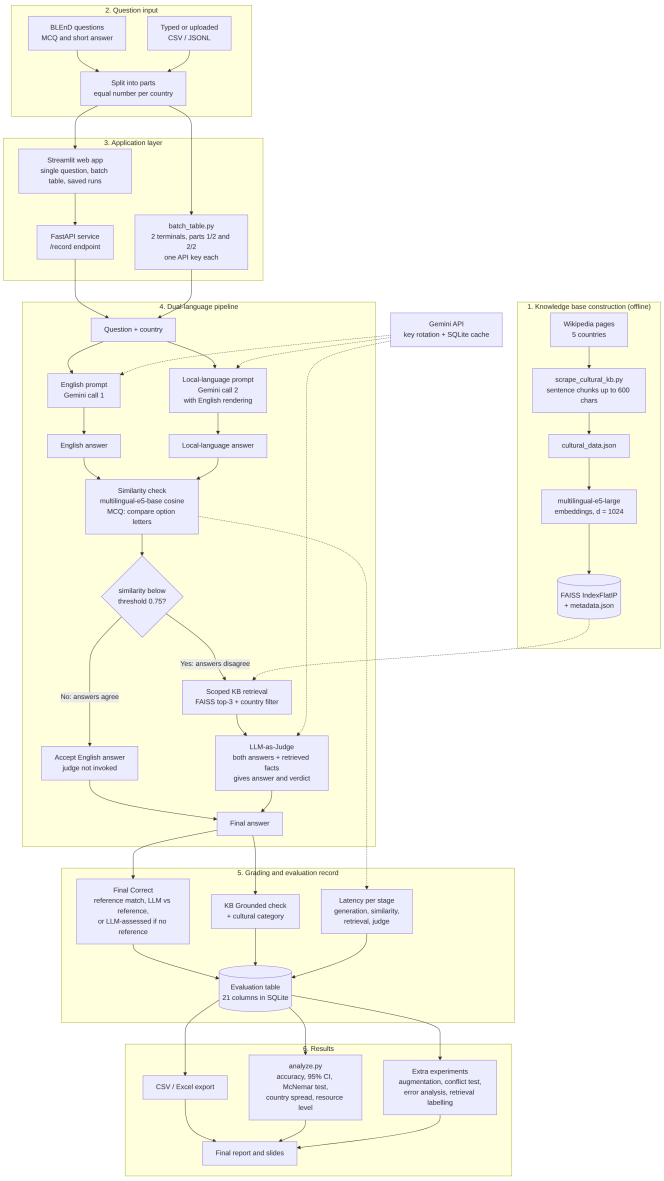


Fig. 1. Dual-language cultural question answering pipeline as implemented: knowledge base construction, question input and splitting, application layer, dual-language generation with a 0.75 similarity threshold, scoped retrieval, LLM-as-Judge, grading and the evaluation record table.

TABLE II
MODEL RESULTS (THRESHOLD = 0.75) — PART 2: METRICS

Record ID	Similarity	Agreement	Final Correct
6	0.8251	Agree	Yes
7	0.8385	Agree	Yes
5	0.8654	Agree	Yes
4	0.9153	Agree	Yes
3	0.8899	Agree	Yes
2	0.8498	Agree	Yes
1	0.8656	Agree	Yes

TABLE III
MODEL RESULTS (THRESHOLD = 0.85) — PART 1: DETAILS

Record ID	Country/Region	Cultural Category	Question	Ground Truth
20	China	Family/Customs	What are the red envelopes given at Chinese New Year called?	hongbao
21	Indonesia	Festival/Holiday	What is the Balinese Day of Silence called?	nyepi
22	Indonesia	Clothing	What wax-resist dyed fabric is Indonesia known for?	batik
23	Azerbaijan	Food	What is Azerbaijan's national rice dish?	plov
16	Ethiopia	Food	What is the staple spongy flatbread of Ethiopia?	injera
18	Ethiopia	Festival/Holiday	Which Ethiopian festival celebrates the finding of the True Cross?	meskel
19	China	Festival/Holiday	Which Chinese festival is celebrated by eating mooncakes?	Mid-Autumn Festival
24	Azerbaijan	Music/Dance	What is the classical improvised music tradition of Azerbaijan called?	mugham

TABLE IV
MODEL RESULTS (THRESHOLD = 0.85) — PART 2: METRICS

Record ID	Similarity	Agreement	Final Correct
20	0.8752	Agree	Yes
21	0.9044	Agree	Yes
22	0.9101	Agree	Yes
23	0.8543	Agree	Yes
16	0.8177	Partial	Yes
18	0.8286	Partial	Yes
19	0.8583	Agree	Yes
24	0.8644	Agree	Yes